Section 1.3 — Functions and Relations

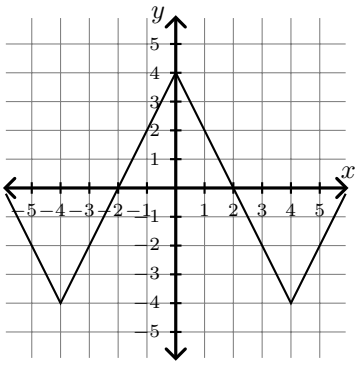
Math 103: College Algebra

1 Understanding Relations

Let $R = \{(2,5),(3,7),(-8,-2),(9,1)\}$. Decide whether each statement below is true.

1. $2R5$
2. $-7R8$
3. $7R3$
4. $-8R-2$

Let S be the relation graphed below. Decide whether each statement below is true.



5. $0S4$
6. $-4S4$
7. $3S-2$
8. $4S-4$

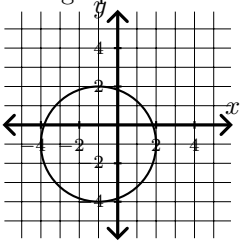
Let $T = \left\{(x,y)|x^2+y^2=1\right\}$. Decide whether each statement below is true.

9. $0T1$
10. $-1T0$
11. $-1T-1$
12. $0.6T-0.8$

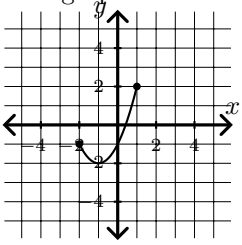
2 Domain and Range of a Relation

For each relation below, describe the domain and range.

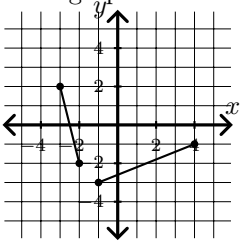
13. $Q = \left\{(2,3),(1,\frac{5}{3}),(-1,35),(\frac{11}{8},2)\right\}$
14. $R = \left\{(1,2),(1,3),(1,4),(1,5)\right\}$
15. $S = \left\{(2,-1),(3,-1),(2,7),(3,9)\right\}$
16. $T = \left\{(11,1),(-2,1),(\pi,1),(2.2,1)\right\}$
17. U is graphed below



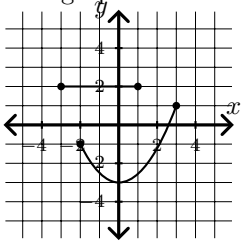
18. V is graphed below



19. W is graphed below



20. Z is graphed below

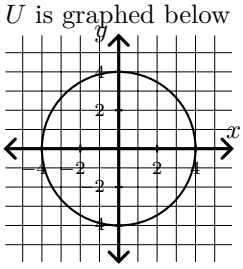


3 Identifying Functions

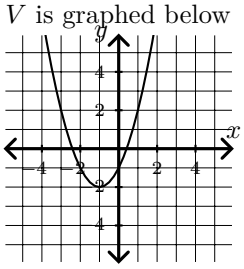
State whether each relation is a function. If not, give two pairs that show why not.

21. $Q = \left\{(1,5),(-3,2),(-1,5),(4,4)\right\}$
22. $R = \left\{(2,3),(2,4)(4,4)\right\}$
23. $S = \left\{(1,1),(2,1),(3,1),(4,1)\right\}$
24. $T = \left\{(-1,-1),(-2,-2),(-1,1),(-2,2)\right\}$

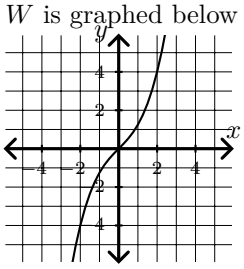
25. U is graphed below



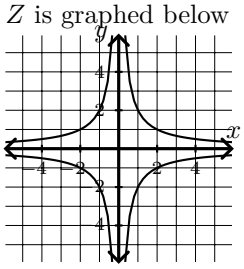
26. V is graphed below



27. W is graphed below



28. Z is graphed below



For each relation below, give two pairs (x, y) that show that the relation is *not* a function. As a hint, remember that x^2 and $(-x)^2$ always give the same (positive) result.

29. $J = \{(x, y) \mid x^2 + y^2 = 8\}$

30. $L = \{(x, y) \mid x = y^2\}$

4 Function Notation

Let $f = \{(-2, 5), (3, -2), (4, 7), (8, 9)\}$. Find each of the following.

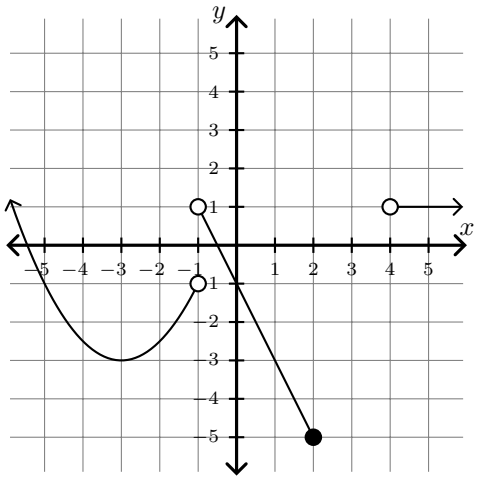
31. $f(4)$

32. $f(-2)$

33. $f(9)$

34. $f(3)$

The function g is graphed below. Find each of the following.



35. $g(-3)$

36. $g(-1)$

37. $g(1)$

38. $g(3)$

5 Equations of Functions

Write each function in relation notation.

39. $f(x) = 2x - 5$

40. $g(x) = x^2 - 7x + 12$

41. $h(x) = \dfrac{3x - 5}{2x + 1}$

Each relation below is a function. Rewrite in function notation. You may need to first solve for y.

42. $R = \{(x, y) \mid y = x^3 - x\}$

43. $S = \{(x, y) \mid 2x - 5y = 10\}$

44. $T = \{(x, y) \mid x^2 - y = 3\}$

Let $k(x) = \dfrac{4x^2 + 2}{x - 7}$. Find each of the following. Give your answer as a fraction.

45. $k(2)$

46. $k(-3)$

47. $k(7)$

48. $k(0)$

49. There is one real number not in the implicit domain of k. Based on your work above, what is $\operatorname{Dom} k$?